FRANMARINE UNDERWATER SERVICES PTY LTD

LOSS OF STEERING

Emergency Procedure

MV INTEGRITY III | UVI: 414562

Marine Order 504 (2024) — Schedule 1 Compliance

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Approval

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Review Frequency

This document shall be reviewed at least every 12 months, or sooner following a significant incident, change of operations, vessel modification, or change in applicable legislation.

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1. PURPOSE

This procedure establishes the actions to be taken in the event of a loss of steering on MV Integrity III.

1.1 Regulatory Reference

Marine Order 504 (2024), Schedule 1, Clause 12.

1.2 Vessel Configuration

MV Integrity III is fitted with twin Suzuki DF175 outboard engines. In the event of a steering system failure, the twin-outboard configuration allows the vessel to be steered using differential thrust — that is, by applying more throttle to one engine than the other to turn the vessel. This is the primary emergency steering method for this vessel.

2. IMMEDIATE ACTIONS

- Reduce speed to idle immediately.
- Alert all crew to the loss of steering.
- Attempt differential thrust steering: increase throttle on the port engine to turn to starboard, or increase throttle on the starboard engine to turn to port. Start with small throttle adjustments to gauge the vessel's response.
- Assess for collision risk. If in a busy waterway or unable to maintain safe directional control, issue a SÉCURITÉ or PAN PAN call on VHF Ch 16 advising of the loss of steering and the vessel's position. Advise the relevant port authority (refer FUS-SMS-INT3-05J for VHF channels).

3. TROUBLESHOOTING

Common causes of steering failure on outboard vessels:

- Low hydraulic steering fluid — check the reservoir level and top up if possible.
- Hydraulic line failure — inspect visible hose runs for leaks or burst hoses.
- Steering cable disconnection — inspect the cable connection at the helm and at the outboard tiller arm.
- Obstruction at the outboard — check for rope, debris, or marine growth around the lower unit preventing the outboard from pivoting.
- Mechanical linkage failure — inspect the tie bar connecting the two outboards (if fitted) for damage or disconnection.

If the fault can be identified and rectified safely, do so. If not, proceed with differential thrust steering to a safe location.

4. EMERGENCY NAVIGATION USING DIFFERENTIAL THRUST

With twin outboards, MV Integrity III can be navigated effectively using differential thrust alone:

- To turn to starboard: increase port engine throttle while reducing or idling the starboard engine.
- To turn to port: increase starboard engine throttle while reducing or idling the port engine.
- To maintain a straight course: match both engine RPMs. Small adjustments will be needed as wind and current act on the vessel.
- Navigate at reduced speed. Differential thrust steering is less precise than hydraulic steering and the vessel will respond more slowly, especially at low speeds.
- Navigate to the nearest safe port, berth, or anchorage.
- If unable to maintain any directional control (e.g. both outboards are locked in one direction), deploy the anchor to arrest drift and prevent grounding or collision. Request a tow if necessary.

5. POST-INCIDENT

- Do not continue normal operations until steering is fully restored and tested.
- Report to the East Coast Operations Superintendent.
- Record all details in the vessel logbook.
- Arrange for repair by a qualified marine mechanic before the vessel returns to service.

6. REFERENCES

- FUS-SMS-INT3-06 Emergency Plan
- FUS-SMS-INT3-07 Maintenance
- Marine Order 504 (2024), Schedule 1, Clause 12